

Utkarsh Singh Yadav

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EDUCATION

Vellore Institute of Technology, Bhopal

B.Tech, Computer Science (Core) — CGPA: 8.53

Expected 2026

Bhopal, M.P.

DAV Centenary Public School, Malerkotla

Class XII & X — 76% / 91%

2020 – 2022

Malerkotla, Punjab

TECHNICAL SKILLS

Languages : Java, Python, SQL

Software Engineering : Data Structures & Algorithms, OOP, REST APIs, Flask, Git/GitHub, MySQL

ML / Computer Vision : Pandas, NumPy, Scikit-learn, OpenCV, Ultralytics (YOLOv8), MediaPipe, Deep Learning, Image Processing

Tools : Git, GitHub, Excel, MySQL, IoT/Embedded fundamentals

CERTIFICATIONS

- **Applied Machine Learning in Python** — University of Michigan (Coursera), Dec. 2025 (Credential: FM9HR9QKZZVR)
- **Introduction to Internet of Things [Elite]** — IIT Kharagpur (NPTEL), Jan.–Apr. 2026 — scored **86% (Elite)**
- **Cloud Computing** — IIT Kharagpur (NPTEL), Jan.–Apr. 2025 — scored 67%

EXPERIENCE

Devplex Technologies Pvt. Ltd.

May 2026 – Jul. 2026

AI/ML Engineer Intern

Bengaluru, Karnataka

- Collaborated daily with senior engineers and cross-functional stakeholders in an **agile team** to scope, prioritize, and ship Python-based computer vision and ML features — translating technical constraints into sprint-level decisions
- **Owned data-processing and model-inference scripts end-to-end**; ran structured root-cause analysis on integration bugs and documented fixes, cutting resolution time and keeping the team aligned via Git-based code review

PROJECTS

Real-Time Face & ID Scanning System | Python, OpenCV, MySQL, QR Code Processing, ML

2024

- Identified manual attendance logging as the core user pain point, then designed and shipped a **full-stack face-recognition and QR-based automation**, replacing the manual process end-to-end
- Designed the schema and query layer balancing speed vs. data integrity, **cutting attendance-lookup time from manual minutes to sub-second** automated retrieval — a measurable, user-facing outcome

Pothole Detection System | Python, YOLOv8 (Ultralytics), OpenCV, Deep Learning

2025

- Framed road-safety monitoring as a smart-city and vehicle-navigation problem, then engineered a **real-time object-detection pipeline (YOLOv8)** performing frame-by-frame video analysis with bounding-box overlays
- Ran iterative trade-off analysis between **inference latency and detection accuracy** to make the model deployable on resource-constrained edge hardware — a feasibility call directly analogous to a build-vs-ship product decision

Hand Sign Recognition System | Python, OpenCV, MediaPipe, Scikit-learn

2024

- Independently scoped and built an **end-to-end gesture-recognition prototype** for assistive-technology and touchless-interface use cases, from problem framing through model selection to a working real-time application
- Validated robustness against real-world variability (lighting, hand orientation) by testing across conditions before considering the prototype 'done' — an early-stage user-validation mindset

EXTRACURRICULAR & ACHIEVEMENTS

- Identified medication non-adherence as a user problem and built **MedBuddy**, a full-stack medication-reminder and social platform with notifications, purchase suggestions, and a gamified reward system to drive behavior change